

Chapter 6: Discrete Probability Distributions

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Introduction

In the previous chapters, we discussed that when the values of a variable can be counted, it is called **discrete** (coin toss, dice roll, the number of students in a classroom).

When a variable can take any range of values, then it is known as **continuous** (temperature, distance, bank balance).

In this chapter, we will be focused on mapping out the **probability distribution** of discrete variables. The two important distributions will be the **Binomial** and **Poisson** distributions.

Probability Distribution

Consider the case of a fair coin toss. We know that the sample space is described by the two outcomes:

$$S = \{H, T\}$$

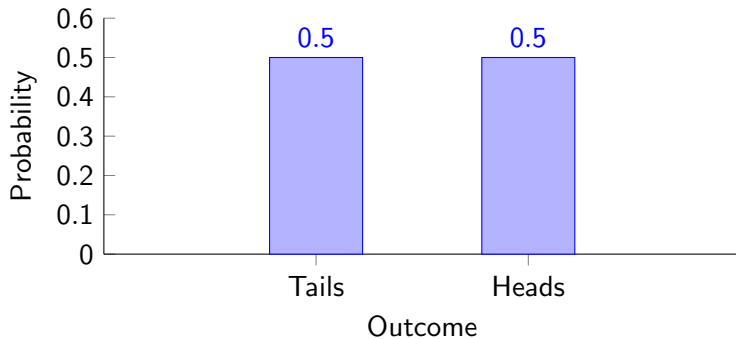
and its associated probabilities were:

$$P(H) = \frac{1}{2} \quad P(T) = \frac{1}{2}$$

Probability Distribution

Graphically, we can illustrate the probability in a graph:

Figure 1: Probability Distribution for One Fair Coin Toss



Probability Distribution

Notice that we are using qualitative notation for heads H , tails T . In this chapter we are interested in using a general quantitative notation so we can incorporate all kinds of examples.

Suppose we toss the coin again, but this time we define the **random variable** X as:

$$X = \begin{cases} 0 & \text{if Tails occurs,} \\ 1 & \text{if Heads occurs.} \end{cases}$$

Probability Distribution

that is, the researcher simply notates 0 as the coin showing tails and 1 if the coin shows heads. This means X can take only two values:

$$X = \{0, 1\}$$

and its associated probabilities are:

$$P(X = 0) = \frac{1}{2} \quad P(X = 1) = \frac{1}{2}$$

Probability Distribution

The way you read this is that the probability that the random variable X is equal to 0 (meaning that it is tails) is $\frac{1}{2}$.

Similarly, the probability that the random variable X is equal to 1 (meaning that is heads) is $\frac{1}{2}$.

Probability Distribution

We can more easily read this in a table:

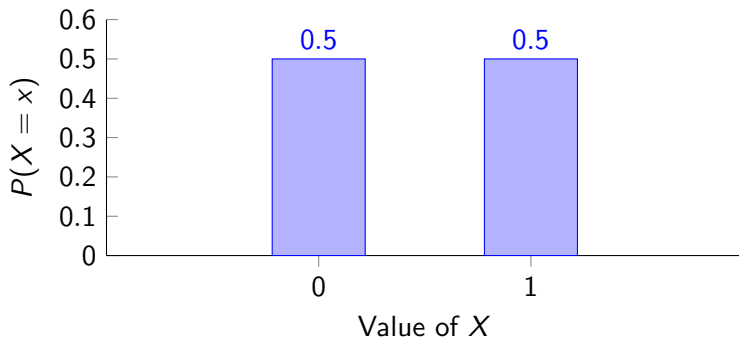
x	$P(X = x)$
0	0.5
1	0.5

This is exactly the same coin toss experiment as before. The difference is that we have now assigned a numerical value to each possible outcome.

Probability Distribution

We can illustrate the **probability distribution** of the random variable X :

Figure 2: Probability Distribution of X , where $X = 0$ for Tails and $X = 1$ for Heads



Probability Distribution

Let us now extend the example where we toss two coins. We know that the sample space will now become:

$$S = \{HH, HT, TH, TT\}$$

Let the random variable X be described as **the number of heads** that the coins show.

Probability Distribution

For each possible outcome, we can see that the number of heads will be:

$$TT \rightarrow 0$$

$$HT \rightarrow 1$$

$$TH \rightarrow 1$$

$$HH \rightarrow 2$$

Probability Distribution

Formally, the random variable X can take the following values:

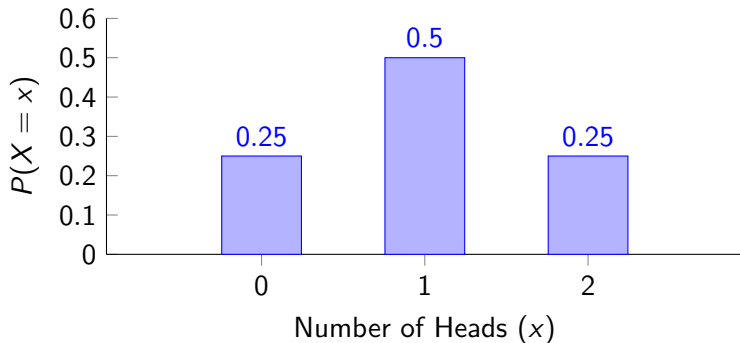
$$X = \{0, 1, 2\}$$

and based on its frequency, the associated probabilities of each value is:

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{2}{4}, \quad P(X = 2) = \frac{1}{4}$$

Probability Distribution

Figure 3: Probability Distribution of the Number of Heads from Two Coin Tosses



Probability Distribution

Activity:

Consider the case of tossing three coins.

- (a) Define the sample space S .
- (b) Let X represent the number of heads the coins show. Describe the frequency of X for each outcome.
- (c) What are the possible values that X can take?
- (d) Based on its frequency, what are the associated probabilities of each value of X ?
- (e) Illustrate the probability distribution of X .

Probability Distribution

Solution:

(a) When three coins are tossed, there are $2^3 = 8$ possible outcomes. The sample space is:

$$S = \{(H, H, H), (H, H, T), (H, T, H), (H, T, T), \\ (T, H, H), (T, H, T), (T, T, H), (T, T, T)\}.$$

Probability Distribution

(b) Let X represent the **number of heads** observed when the three coins are tossed. We can determine the value of X associated with each outcome:

Outcome	Number of Heads (X)
(H, H, H)	3
(H, H, T)	2
(H, T, H)	2
(H, T, T)	1
(T, H, H)	2
(T, H, T)	1
(T, T, H)	1
(T, T, T)	0

Probability Distribution

(c) The possible values that the random variable X can take are:

$$x = 0, 1, 2, 3.$$

That is, when three coins are tossed, we can observe zero, one, two, or three heads.

Probability Distribution

(d) Since the coins are fair, all eight outcomes in the sample space are equally likely. We can therefore calculate the probability associated with each value of X using:

$$P(X = x) = \frac{\text{Frequency of } x}{\text{Total number of possible outcomes}}.$$

Probability Distribution

Therefore:

$$P(X = 0) = \frac{1}{8} = 0.125,$$

$$P(X = 1) = \frac{3}{8} = 0.375,$$

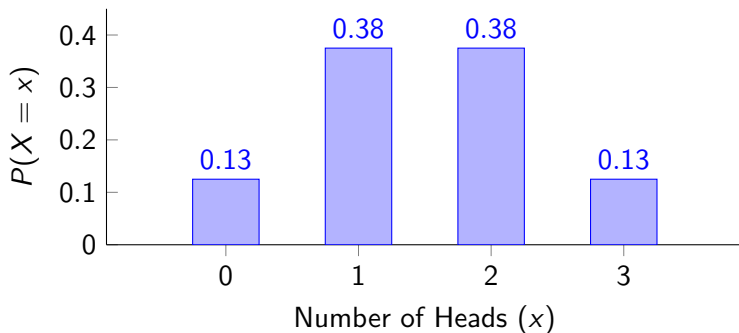
$$P(X = 2) = \frac{3}{8} = 0.375,$$

$$P(X = 3) = \frac{1}{8} = 0.125.$$

Probability Distribution

(e) The probability distribution of X can be illustrated graphically:

Figure 4: Probability Distribution of the Number of Heads from Three Coin Tosses



The Mean and the Expectations Operator

In the previous chapter, we introduced the **mean** as a measure of central tendency. We can apply a similar idea to a probability distribution using the concept of an **expected value**.

The Mean and the Expectations Operator

Consider the example of rolling a fair six-sided die. Let the random variable X represent the number shown on the die. The possible values of X are:

$$x = 1, 2, 3, 4, 5, 6.$$

Since the die is fair, each value has the same probability:

$$P(X = x) = \frac{1}{6}.$$

The probability distribution of X is therefore:

The Mean and the Expectations Operator

x	$P(X = x)$
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\frac{1}{6}$
4	$\frac{1}{6}$
5	$\frac{1}{6}$
6	$\frac{1}{6}$

The Mean and the Expectations Operator

To find the mean of a probability distribution, we multiply each possible value of X by its associated probability and add the results together.

In general, the **expected value** of a random variable is:

$$E(X) = \sum xP(X = x)$$

where E is known as the **expectation operator**.

The Mean and the Expectations Operator

For the die example, we can write this as:

$$\begin{aligned}\mu = E(X) &= 1 \left(\frac{1}{6} \right) + 2 \left(\frac{1}{6} \right) + 3 \left(\frac{1}{6} \right) + 4 \left(\frac{1}{6} \right) + 5 \left(\frac{1}{6} \right) + 6 \left(\frac{1}{6} \right) \\ &= \frac{1 + 2 + 3 + 4 + 5 + 6}{6} \\ &= \frac{21}{6} \\ &= 3.5.\end{aligned}$$

The expected value (mean) of the die roll is therefore 3.5.

Variance and Standard Deviation

The **variance** of a discrete random variable measures the expected squared distance of its possible values from the mean.

$$\text{Var}(X) = \sum (x - \mu)^2 P(X = x)$$

Variance and Standard Deviation

For the die example, given that $\mu = 3.5$:

$$\begin{aligned}\text{Var}(X) &= (1 - 3.5)^2 \left(\frac{1}{6}\right) + (2 - 3.5)^2 \left(\frac{1}{6}\right) \\ &\quad + (3 - 3.5)^2 \left(\frac{1}{6}\right) + (4 - 3.5)^2 \left(\frac{1}{6}\right) \\ &\quad + (5 - 3.5)^2 \left(\frac{1}{6}\right) + (6 - 3.5)^2 \left(\frac{1}{6}\right) .\end{aligned}$$

Variance and Standard Deviation

Therefore:

$$\begin{aligned}\text{Var}(X) &= \frac{6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25}{6} \\ &= \frac{17.5}{6} \\ &= 2.917.\end{aligned}$$

Variance and Standard Deviation

The variance is expressed in squared units. As we learned previously, we can take the square root of the variance to obtain the **standard deviation**:

$$\begin{aligned}\sigma &= \sqrt{\text{Var}(X)} \\ &= \sqrt{2.917} \\ &= 1.708.\end{aligned}$$

Binomial Distribution

Now that we understand what probability distributions are, we can work with the famous kinds of discrete probability distributions.

Binomial Distribution

A random variable X is said to follow a **binomial distribution** if it satisfies these four basic conditions:

- 1 There is a fixed number of trials n .
- 2 Each trial has only two possible outcomes, 0 or 1 (This is conventionally called failure or success.).
- 3 The probability of the outcome occurring is a constant π .
- 4 The trials n are independent.

Binomial Distribution

If this is satisfied, we denote the random variable X as:

$$X \sim \text{Bin}(n, \pi)$$

Binomial Distribution

Activity:

Consider the experiment of tossing a two coins. Explain how this generates a random variable X that is drawn from a binomial distribution.

Binomial Distribution

Solution:

The coin toss is either $X = 1$ if heads or $X = 0$ if tails. This experiment satisfies the binomial distribution conditions:

- 1 There is a fixed number of trials. Tossing two coin means $n = 2$.
- 2 Each trial only has two possible outcomes, heads or tails ($X = 1$ or 0).
- 3 The probability of seeing the event is fixed. This is true because $\pi = 0.5$ in all cases.
- 4 The trials are independent. The coins being tossed have nothing to do with each other.

Binomial Distribution

When a random variable satisfies the binomial distribution conditions, then we can apply a general formula to extract the probability of X showing a particular number x :

$$P(X = x) = nC_x \cdot \pi^x (1 - \pi)^{n-x}$$

The nC_x component is simply the combination rule that we discussed in the previous chapter, but we instead use the values for n and x . The π component represents the probability of the event occurring.

Binomial Distribution

Lets consider again the basic example of two coin tosses. We know that the sample space is:

$$S = \{HH, HT, TH, TT\}$$

If the random variable X measures the number of heads, then we know that the probability distribution is represented by (Read section 6.2 if you misremember this part.):

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{2}{4}, \quad P(X = 2) = \frac{1}{4}$$

Binomial Distribution

However since the coin toss generates an the random variable $X \sim \text{Bin}(2, 0.5)$, we can instead just input the values for n and x to recover this answer.

For the formula:

$$P(X = x) = nC_x \cdot \pi^x (1 - \pi)^{n-x}$$

Given that $X = \{0, 1, 2\}$ and $X \sim \text{Bin}(2, 0.5)$, we have:

Binomial Distribution

$$n = 2,$$

$$\pi = 0.5,$$

$$1 - \pi = 0.5.$$

For $X = 0$:

Binomial Distribution

$$\begin{aligned}P(X = 0) &= 20(0.5)^0(1 - 0.5)^{2-0} \\&= 1(1)(0.5)^2 \\&= 0.25.\end{aligned}$$

For $X = 1$:

Binomial Distribution

$$\begin{aligned}P(X = 1) &= {}^2C_1(0.5)^1(1 - 0.5)^{2-1} \\&= 2(0.5)(0.5) \\&= 0.50.\end{aligned}$$

For $X = 2$:

Binomial Distribution

$$\begin{aligned}P(X = 2) &= {}^2C_2(0.5)^2(1 - 0.5)^{2-2} \\&= 1(0.5)^2(1) \\&= 0.25.\end{aligned}$$

which is identical to our answer from before.

Binomial Distribution

When a random variable is drawn from a binomial distribution, we can define its mean and variance as follows:

$$\text{Mean} = n\pi$$

$$\text{Variance} = n\pi(1 - \pi)$$

Poisson Distribution

Suppose we are interested in the analysing the probability of the below events:

- The number of customers arriving at a BBK branch within 1 hour.
- The number of international calls made in 1 day.
- The number of traffic violations in Riffa in October.

Poisson Distribution

Notice how the events are restricted by hour, day or month.

Basically, when a discrete random variable X represents the number of times an event occurs within a specific interval, we say that X is drawn from a **Poisson distribution**^a.

Formally, we denote this as:

$$X \sim \text{Poisson}(\mu)$$

where μ represents the average number of occurrences of that event within the specified interval.

^aTo be clear, the Poisson distribution can be used for **any interval**, such as time, space, distance, area, volume, etc. For example, we may be interested in the number of defects within one meter of fabric.

Poisson Distribution

Consider an example where a Batelco call centre receives an average 4 calls every 10 minutes. Let us define the random variable X as:

X = Number of calls received in 10 minutes

Even though only 4 calls are observed on average within 10 minutes, it is theoretically possible that X can take an infinite amount of (non-negative integer) values:

$$X = \{0, 1, 2, 3, 4, 5, 6, 7, 8, \dots\}$$

Poisson Distribution

If the Batelco manager believes that X is drawn from a Poisson distribution, then it they can calculate the probability of any number of calls that might happen within the 10 minute period given the average.

The probability of observing exactly $X = x$ occurrences is:

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

Poisson Distribution

For example, the Batelco manager might be interested in calculating the probability of seeing 7 calls within 10 minutes **given that** the average is 4. In this case:

$$P(X = 7) = \frac{\exp^{-4} 4^x}{7!}$$

Poisson Distribution

Under general conditions, it is possible to show that the Poisson distribution has an identical mean and variance as described below:

$$\mu = n\pi$$

$$\sigma^2 = n\pi$$

Notice how this is the same as the mean of the binomial distribution.